

```

typedef struct {
    unsigned char boot_type; // 0x80 -
Inactive; 0x80 - Active (Bootable)
    unsigned char start_head;
    unsigned char start_sec;6;
    unsigned char start_cyl_hi;2;
    unsigned char start_cyl;
    unsigned char part_type;
    unsigned char end_head;
    unsigned char end_sec;6;
    unsigned char end_cyl_hi;2;
    unsigned char end_cyl;
    unsigned long abs_start_sec;
    unsigned long sec_in_part;
} PartEntry;

typedef struct {
    unsigned char boot_code[MBR_DISK_
SIGNATURE_OFFSET];
    unsigned long disk_signature;
    unsigned short pad;
    unsigned char pt[PARTITION_TABLE_
SIZE];
    unsigned short signature;
} MBR;

void print_computed(unsigned long sector)
{
    unsigned long heads, cyls, tracks,
sectors;

    sectors = sector % 63 + 1 /* As
indexed from 1 */;
    tracks = sector / 63;
    cyls = tracks / 255 + 1 /* As indexed
from 1 */;
    heads = tracks % 255;
    printf("%3d/%5d/%3d", heads, cyls,
sectors);
}

int main(int argc, char *argv[]) {
    char *dev_file = "/dev/sda";
    int fd, i, rd_val;
    MBR m;
    PartEntry *p = (PartEntry *)(&m.pt);

    if (argc == 2) {
        dev_file = argv[1];
    }
    if ((fd = open(dev_file, O_RDONLY)) ==
-1) {
        fprintf(stderr, "Failed opening

```

```

%s: ", dev_file);
        perror("");
        return 1;
    }
    if ((rd_val = read(fd, &n,
sizeof(m))) != sizeof(m)) {
        fprintf(stderr, "Failed reading
%s: ", dev_file);
        perror("");
        close(fd);
        return 2;
    }
    close(fd);
    printf("VNDOS type Partition Table of
%s:\n", dev_file);
    printf(" B Start (H/C/S) End
(H/C/S) Type StartSec TotSec\n");
    for (i = 0; i < 4; i++) {
        printf("%d:%d (%3d/%4d/%2d)
(%3d/%4d/%2d) %8X %10d %9d\n",
            i + 1, !!p[i].boot_type &
0x80),
            p[i].start_head,
            1 + ((p[i].start_cyl_hi < 8)
| p[i].start_cyl),
            p[i].start_sec,
            p[i].end_head,
            1 + ((p[i].end_cyl_hi < 8) |
p[i].end_cyl),
            p[i].end_sec,
            p[i].part_type,
            p[i].abs_start_sec, p[i].
sec_in_part);
    }
    printf("\nRe-computed Partition Table
of %s:\n", dev_file);
    printf(" B Start (H/C/S) End
(H/C/S) Type StartSec TotSec\n");
    for (i = 0; i < 4; i++) {
        printf("%d:%d ", i + 1, !!p[i].
boot_type & 0x80));
        print_computed(p[i].abs_start_
sec);
        printf(" ");
        print_computed(p[i].abs_start_sec
+ p[i].sec_in_part - 1);
        printf(" %02X %10d %9d\n", p[i].
part_type,
            p[i].abs_start_sec, p[i].
sec_in_part);
    }
    printf("\n");
    return 0;
}

```

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